

RONNAKRIT RATTANASRIAMPAIPONG

POSTDOCTORAL RESEARCH ASSOCIATE

Department of Geosciences, The University of Arizona

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EDUCATION

TEXAS A&M UNIVERSITY

Ph.D. in Oceanography (Paleoceanography/Organic Geochemistry)

College Station, TX

2018–2023

CHULALONGKORN UNIVERSITY

M.S. in Petroleum Geoscience (International Program)

B.S. in Geology

Bangkok, Thailand

2015–2016

2009–2013

PROFESSIONAL APPOINTMENTS

THE UNIVERSITY OF ARIZONA, DEPARTMENT OF GEOSCIENCES

Postdoctoral Research Associate

NOAA Climate & Global Change Postdoctoral Fellow

Tucson, AZ

2026–Present

2023–2025

TEXAS A&M UNIVERSITY, DEPARTMENT OF OCEANOGRAPHY

Graduate Research Assistant

Graduate Assistant — Teaching (*teaching assistant*)

Graduate Assistant Lecturer (*instructor of record*)

College Station, TX

2018–2023

2023

2021–2022

CHEVRON THAILAND EXPLORATION AND PRODUCTION, LIMITED

Development Geologist, Satun-Funan Asset Team

Horizons Earth Scientist, Reservoir Management Team

Undergraduate Student Intern, Reservoir Management Team

Bangkok, Thailand

2016–2018

2013–2016

Summer 2012, 2013

GRANTS & FELLOWSHIPS

Total research and fellowship funding awarded: \$326,000.

EXTERNAL

NOAA Climate & Global Change Postdoctoral Fellowship

\$200,000 over two years; up to 8 fellows selected worldwide.

2023–2025

Schlanger Ocean Drilling Fellowship, NSF/U.S. Science Support Program

\$30,000 for one year; up to 8 fellows selected nationwide.

2021–2022

Fulbright Thai Graduate Scholarship

\$70,000 over two years; up to 7 scholars selected nationwide.

2018–2020

Texas Sea Grant Grants-In-Aid Graduate Research Grant

\$2,000 over two years; competitive state-wide award, NOAA Sea Grant College Program.

2022–2024

INTERNAL

Texas A&M Dissertation Fellowship

\$24,000 for one year; up to 15 fellows selected university-wide.

2023–2024

PUBLICATIONS

PAPERS PUBLISHED IN REFEREED JOURNALS

145 total citations, h-index 4 | [google scholar](#) 

2026

7. Elling, F. J., Inglis, G. N., Davtian, N., Hurley, S. J., Naafs, B. D. A., O'Connor, L. K., Pancost, R. D., Peterse, F., Rattanasriampaipong, R., Rice, A., Schubotz, F., Damsté, J. S. S., Wörmer, L., Auderset, A., Becker, K. W., Berke,

M. A., Bijl, P. K., Jonge, C. D., Dong, L., Duncan, B. J., Jones, T. D., Fenies, P., Fietz, S., Ho, S. L., Huguët, A., Huguët, C., Kaiser, J., Liu, X.-L., Mollenhauer, G., Naeher, S., Pearson, A., Śliwińska, K. K., Sluijs, A., de Oca, C. S.-M., Tian, Q., Tierney, J. E., Varma, D., Xiao, W., Zhang, Y. G., and Zhao, X. (2026). Archaeal tetraether lipids as tracers for past marine environmental change. *Paleoceanography and Paleoclimatology*, 41:e2025PA005373. <https://doi.org/10.1029/2025PA005373> Contributing author. Drafted the review's synthesis of the biological sources of marine tetraether lipids, distributed across several sections of the final paper.

6. Tran, T. T., Tierney, J. E., Holbourn, A., and **Rattanasriampaipong, R.** (2026). Biomarker constraints on indo-pacific warm pool temperature during the late pleistocene: The role of orbital forcing. *Paleoceanography and Paleoclimatology*, 41:e2025PA005276. <https://doi.org/10.1029/2025PA005276>

2025

5. Bijl, P. K., Śliwińska, K. K., Duncan, B., Huguët, A., Naeher, S., **Rattanasriampaipong, R.**, Sosa-Montes de Oca, C., Auderset, A., Berke, M. A., Kim, B. S., Davtian, N., Dunkley Jones, T., Eefting, D. D., Elling, F. J., Fenies, P., Inglis, G. N., O'Connor, L., Pancost, R. D., Peterse, F., Rice, A., Sluijs, A., Varma, D., Xiao, W., and Zhang, Y. G. (2025). Reviews and syntheses: Best practices for the application of marine GDGTs as proxy for paleotemperatures: sampling, processing, analyses, interpretation, and archiving protocols. *Biogeosciences*, 22(21):6465–6508. <https://doi.org/10.5194/bg-22-6465-2025>. Contributing author. Led the writing of the 'data reporting and archiving' section and produced Figures 1 and 4.

4. **Rattanasriampaipong, R.**, Tierney, J. E., Abell, J. T., and Gilmore, L. D. (2025). A nutrient effect on the TEX₈₆ paleotemperature proxy. *Geophysical Research Letters*, 52(12):e2025GL115237. <https://doi.org/10.1029/2025GL115237>

2022

3. Judd, E. J., Tierney, J. E., Huber, B. T., Wing, S. L., Lunt, D. J., Ford, H. L., Inglis, G. N., McClymont, E. L., O'Brien, C. L., **Rattanasriampaipong, R.**, Si, W., Staitis, M. L., Thirumalai, K., Anagnostou, E., Cramwinckel, M. J., Dawson, R. R., Evans, D., Gray, W. R., Grossman, E. L., Henehan, M. J., Hupp, B. N., MacLeod, K. G., O'Connor, L. K., Sánchez Montes, M. L., Song, H., and Zhang, Y. G. (2022). The PhanSST global database of Phanerozoic sea surface temperature proxy data. *Scientific Data*, 9(1). <https://doi.org/10.1038/s41597-022-01826-0>. Contributing author. Entered and quality-controlled a substantial share of the compiled records.
2. **Rattanasriampaipong, R.**, Zhang, Y. G., Pearson, A., Hedlund, B. P., and Zhang, S. (2022). Archaeal lipids trace ecology and evolution of marine ammonia-oxidizing archaea. *Proceedings of the National Academy of Sciences*, 119(31):e2123193119. <https://doi.org/10.1073/pnas.2123193119>

2016

1. **Rattanasriampaipong, R.** (2016). Potential sources of mercury in Southern Pattani Basin, the Gulf of Thailand. *Bulletin of Earth Sciences of Thailand*, 8(2 SE - Research Articles):133–144. <https://doi.org/10.13140/RG.2.2.33243.67369>

MANUSCRIPT(S) SUBMITTED, UNDER REVIEW, UNDER REVISION, PREPRINT AND/OR IN-PRESS

3. Zhang, W., Tang, X., Yao, W., Qian, J., He, W., Zheng, Y., Zhu, Y., **Rattanasriampaipong, R.**, Chen, Y., Zheng, F., and Zhang, C. (2026). Enhanced production and methylation of branched GDGTs responding to water content-mediated redox changes in Tibetan Plateau permafrost soils. *EGUsphere* [preprint]. <https://doi.org/10.5194/egusphere-2026-3837>
2. **Rattanasriampaipong, R.**, Tierney, J. E., Elling, F. J., and Inglis, G. N. (2026). TEXAS: A proxy system model for TEX₈₆ paleothermometry. *ESS Open Archive*, 2026. <https://doi.org/10.22541/essoar.15002926/v1>
1. **Rattanasriampaipong, R.**, Zhang, Y. G., Alo, O., Liu, X.-L., Zhang, Y., Kim, B., Marcantonio, F., Bassinot, F., and Li, T. (2024). Methylation index of Overly Branched tetraether lipids (MOB): a proxy for deep ocean (de)oxygenation? *ESS Open Archive*. <https://doi.org/10.22541/essoar.171415886.62339907/v1>

TECHNICAL REPORTS

1. **Recommended Location, Logging and Coring Program Geological Reports**, Chevron Thailand, 2013–2018. Full pre-drill assessments for four drilling projects, including geological risk, economic justification, and wireline logging and formation testing programs.

SOFTWARE & OPEN-SOURCE CONTRIBUTIONS

TEXAS (TetraEther indeX of Ammonia oxidizerS) — Hierarchical Bayesian proxy system model for TEX₈₆ paleothermometry, implemented in Stan. Linked to **Rattanasriampaipong, R.**, Tierney, J. E., Elling, F. J., and Inglis, G. N. (2026). TEXAS: A proxy system model for TEX₈₆ paleothermometry. *ESS Open Archive*, 2026.

github.com/PaleoLipidRR/TEXAS

Analysis code: Nutrient Effect on TEX₈₆ — **Rattanasriampaipong, R.**, Tierney, J. E., Abell, J. T., and Gilmore, L. D. (2025). A nutrient effect on the TEX₈₆ paleotemperature proxy. *Geophysical Research Letters*, 52(12):e2025GL115237.

github.com/PaleoLipidRR/nutrient-effect-on-TEX

Analysis code: GDGT-2/GDGT-3 traces AOA Ecology and Evolution — **Rattanasriampaipong, R.**, Zhang, Y. G., Pearson, A., Hedlund, B. P., and Zhang, S. (2022). Archaeal lipids trace ecology and evolution of marine ammonia-oxidizing archaea. *Proceedings of the National Academy of Sciences*, 119(31):e2123193119.

github.com/PaleoLipidRR/marine-AOA-GDGT-distribution

TECHNICAL SKILLS & LANGUAGES

Languages: Thai (Native), English (Full professional proficiency)

Programming & Statistical Software: Python (Proficient), MATLAB (Intermediate), Stan (Intermediate)

Laboratory: Accelerated solvent extraction, silica-gel column chromatography, LC-MS lipid biomarker analysis

Data Visualization & Analysis: Tableau, Spotfire, Python visualization libraries

Documentation & Design: LaTeX/Overleaf, MS Office Suite, Adobe Illustrator, Inkscape

RESEARCH EXPERIENCE

POSTDOCTORAL RESEARCH, DEPARTMENT OF GEOSCIENCES, THE UNIVERSITY OF ARIZONA

2023–Present

Mentor: Jessica E. Tierney

Title: *TEXAS: Towards a full proxy system modeling of TetraEther indeX of Ammonia oxidizerS and reanalysis of temperature trends for the past 100 million years*

- Designed and developed a comprehensive proxy system modeling framework for the TEX₈₆ paleotemperature proxy to reconstruct ancient climate conditions
- Discovered and quantified the influence of nutrient levels on TEX₈₆ proxy accuracy, identifying “nutrient stress” effects in surface marine sediments through analysis of global and regional lipid distribution patterns
- Developing an innovative lipid-based paleotemperature proxy using Bayesian inference methodology implemented in Stan programming language

DOCTORAL RESEARCH, DEPARTMENT OF OCEANOGRAPHY, TEXAS A&M UNIVERSITY

2018–2023

Committee: Yige Zhang (Chair), Ethan Grossman, Robert Korty, Jason Sylvan

Collaborators & Mentors: Ann Pearson (Harvard), Brian Hedlund (UNLV), Shuang Zhang (TAMU)

“Beyond TEX₈₆: Evaluating Archaeal Evolution Coupled with Oceans and Climate Changes using Tetraether Lipids”

- Compiled and synthesized a large archaeal membrane lipid database (N>3000) spanning 192 million years, and identified the spatio-temporal gaps that thin the record before ca. 100 Ma
- Characterized archaeal lipid distribution patterns linked to archaeal ecology in marine environments
- Investigated potential of novel subclasses of archaeal membrane lipids as a proxy for ocean (de)oxygenation

MASTER’S THESIS, DEPARTMENT OF GEOLOGY, CHULALONGKORN UNIVERSITY

2016

Advisor: John Warren

“Potential Sources of Mercury in Southern Pattani Basin, the Gulf of Thailand”

UNDERGRADUATE SENIOR YEAR RESEARCH, DEPARTMENT OF GEOLOGY, CHULALONGKORN UNIVERSITY

2011–2013

Advisor: Kruawun Jankaew

“Quantification of Tsunami Magnitude from Sedimentation Modeling of Re-occurring Indian Ocean Tsunamiites at Phra Thong Island, Phang Nga, Thailand”

FIELD & SHIPBOARD EXPERIENCE

GRADUATE RESEARCH STAFF ON R/V TRIDENT, GALVESTON BAY, TEXAS A&M UNIVERSITY 2018

- Conducted oceanographic sampling operations on a day cruise in Galveston Bay: Niskin-bottle and handheld CTD seawater sampling, seafloor grab sampling, Secchi-disk turbidity, and seawater pumping for biological and trace-metal analyses.

SHORE-BASED OPERATIONAL GEOLOGIST, CHEVRON THAILAND EXPLORATION AND PRODUCTION LIMITED 2017

- Achieved 90% accuracy in pre-drill vs. post-drill hydrocarbon reserves estimation across 20 gas wells.
- Performed real-time stratigraphic correlation using wireline triple-combo logs and reassessed pore-pressure profiles during drilling.

WELLSITE GEOLOGIST TRAINEE, CHEVRON THAILAND EXPLORATION AND PRODUCTION LIMITED 2014

- Monitored real-time drilling operations in the Gulf of Thailand for two weeks on a barge tender rig, covering formation pressure acquisition, formation integrity tests (FIT), and rig-floor operations.

UNDERGRADUATE FIELD ASSISTANT, CHULALONGKORN UNIVERSITY Summers 2011–2013

- Conducted multi-season field excavation and stratigraphic analysis of tsunami deposits on Phra Thong Island, Thailand: trench logging, grain-size and sedimentological description, and sample collection for geochemical analysis.

TEACHING EXPERIENCE

Taught and mentored over 700 undergraduate students, the large majority of them non-majors, across four sections as **instructor of record** for an introductory ocean-and-climate course and two courses as a **teaching assistant**. My teaching emphasizes active learning through hands-on data analysis, real-world applications of Earth science, and practices that support students from a wide range of backgrounds and preparation levels.

Course	Role	Term	Students (Rating)**
<i>Department of Oceanography, Texas A&M University</i>			
The Blue Planet — Our Oceans (OCNG 251)	Instructor of record	Spring 2021	194 (4.78/5.00)
	Instructor of record	Summer 2021	48 (4.86/5.00)
	Instructor of record	Summer 2022	116 (4.64/5.00)
	Instructor of record	Fall 2022	36 (4.41/5.00)
	Teaching assistant	Spring 2023	300
Data Analysis Methods in Geosciences (GEOS 470)	Teaching assistant	Spring 2023	30 (5.00/5.00)

** Student evaluations of my own teaching, on the item “*The instructor fostered an effective learning environment*” (Scale: 1 = Strongly Disagree, 5 = Strongly Agree). Ratings are shown only for roles in which students evaluated me directly.

- **OCNG 251, The Blue Planet — Our Oceans**, is Texas A&M’s introductory oceanography course for non-majors and satisfies a general-education science requirement. As instructor of record I held sole responsibility for syllabus design, recorded lectures, assessment, and grading across four asynchronous online sections enrolling up to 194 students. Its central theme is the ocean’s role in the climate system.
- **GEOS 470, Data Analysis Methods in Geosciences**, is an upper-level, project-based course in which students analyze real geoscience datasets in MATLAB; I led programming sessions and one-on-one code review for a class of 30.

Guest lectures and single-session teaching are listed under *Invited Talks & Lectures*.

PROFESSIONAL EXPERIENCE

CHEVRON THAILAND EXPLORATION AND PRODUCTION, LIMITED, BANGKOK, THAILAND

2013–2018

- Conducted detailed geological assessments in extensional basins (fluvio-deltaic environments) using integrated datasets including 3D seismic, wireline logs, and production records.
- Managed and analyzed geological data from over 5,000 wells to generate pore pressure analyses, geostatistical reserve estimates, and regional pressure-temperature models.
- Collaborated across disciplines (geology, drilling, reservoir engineering) to optimize exploration and development strategies in the Gulf of Thailand.
- Developed a novel method for evaluating reservoir pressure using gas shows and wireline log data.
- Contributed to reserves estimation, stratigraphic correlation, and target redefinition for future platforms within the Dara geologic trend.

INVITED TALKS & LECTURES

CONDUCTED IN ENGLISH

- 12/2025 **University of Nevada, Las Vegas, Las Vegas, NV**
Archaeal lipids and ocean paleoecology, Virtual Lecture.
- 4/2025 **University of Colorado Boulder GeoBioSupergroup Meeting, Boulder, CO**
Disentangling nutrient effects on TEX₈₆ paleotemperature proxy, Virtual Talk.
- 10/2023 **50th SACNAS NDiSTEM, Portland, OR**
Archaeal lipids trace ecology and evolution of marine ammonia-oxidizing archaea, Talk.
- 9/2023 **2nd International GDGT Workshop, Virtual, ETH Zurich, Switzerland**
Marine AOA ecology shifts with Earth's climate, Talk. [[Slides](#)]
- 10/2022 **Southern University of Science and Technology (SUSTech), China**
Archaeal lipids reveal distinct AOA ecology in past warm oceans, Lecture Series on Modern Geobiology. [[Slides](#)]
- 10/2022 **Lamont-Doherty Earth Observatory (LDEO), Columbia University, NY**
A suppression of deep-water clades of marine ammonia oxidizers in past warm oceans, Biology and Paleo Environment (BPE) Seminar Series, Talk.
- 10/2022 **Pal(a)eo Early Career Seminar (Pal(a)eoPERCS), Virtual**
Untapped potential of archaeal lipids beyond ocean temperature reconstructions, Talk. [[Recording](#)]
- 7/2022 **U.S. Advisory Committee for Scientific Ocean Drilling (USAC), Summer Meeting, Virtual**
Invited Talk on marine geochemistry and scientific ocean drilling.
- 12/2021 **Chulalongkorn University, Marine Science Department, Thailand**
The Anthropocene: Human Footprints on Planet Earth, Lecture. [[Slides](#)]

CONDUCTED IN THAI

- 6/2025 **Early Career Ocean Professionals (ECOP) Thailand, Virtual**
Identifying nutrient stress on TEX₈₆ paleotemperature proxy, Talk. [[Slides](#)]
- 1/2023 **Kasetsart University, Department of Earth Sciences, Thailand**
Ammonia Oxidizers in Past Warm Oceans, Lecture. [[Slides](#)]
- 10/2022 **Burapha University, Department of Marine Technology, Virtual**
Inferred paleoecology of marine archaea from today's oceans, Marine Technology Colloquium #62. [[Recording](#)]
- 2/2022 **Chulalongkorn University, Department of Geology, Virtual**
Life after Resignation: 'Fulbright' has So Much to Offer, Talk. [[Slides](#)]
- 9/2020 **Burapha University, Department of Marine Technology, Virtual**
Fossil lipids: Thermometers for the Earth's Climate History, Talk. [[Recording](#)]

PRESENTATIONS

ACADEMIC & SCIENTIFIC CONFERENCES

14. **Rattanasriampaipong, R.**, Tierney, J. E., Elling, F., and Inglis, G. N. (2025, December). *The PETM is not that hot in TEXAS: Reconstructing hyperthermals with a proxy system model for tetraether lipids*. Poster presentation at the 2025 AGU Fall Meeting, New Orleans Ernest N. Morial Convention Center, New Orleans, LA, USA.
13. **Rattanasriampaipong, R.**, Tierney, J. E., Abell, J. T., and Gilmore, L. D. (2025, September). *Disentangling nutrient effect on TEX₈₆ paleotemperature proxy*. Oral presentation at the 2025 International Meeting on Organic Geochemistry, Hilton Porto Gaia, Vila Nova de Gaia, Portugal.
12. **Rattanasriampaipong, R.**, Tierney, J. E., Elling, F., and Inglis, G. N. (2024, December). *Rethinking TEX₈₆ temperature calibration with TEXAS-PSM*. Oral presentation at the 2024 AGU Fall Meeting, Walter E. Washington Convention Center, Washington, D.C., USA.
11. **Rattanasriampaipong, R.** and Tierney, J. E. (2024, August). *TEXAS-PSM: Towards a full proxy system modeling of TetraEther index of Ammonia oxidizerS*. Poster presentation at the 2024 Gordon Research Conference on Organic Geochemistry, Holderness School, Holderness, NH, USA.
10. **Rattanasriampaipong, R.** and Tierney, J. E. (2024, July). *TEXAS-PSM: Towards a full proxy system modeling of TetraEther index of Ammonia oxidizerS*. Poster presentation at the 2024 Gordon Research Seminar on Organic Geochemistry, Holderness School, Holderness, NH, USA.
9. **Rattanasriampaipong, R.** and Tierney, J. E. (2024, July). *Rethinking proxy calibration framework for TEX₈₆ paleothermometry*. Oral presentation at the 16th NOAA Climate and Global Change Postdoctoral Fellowship Summer Institute, Holiday Inn, Steamboat Springs, CO, USA.
8. **Rattanasriampaipong, R.**, Zhang, Y. G., Alo, O., Liu, X. L., Zhang, Y., Kim, B., Marcantonio, F., and Bassinot, F. (2023, December). *Bacterial tetraether lipids as a proxy for ocean (de)oxygenation*. Abstract (PP33B-04) presented at the 2023 AGU Fall Meeting, Moscone Center, San Francisco, CA, USA.
7. **Rattanasriampaipong, R.**, Zhang, Y. G., Pearson, A., Hedlund, B., and Zhang, S. (2022, December). *Archaeal lipids suggest ecological shifts of marine ammonia-oxidizing archaea in greenhouse worlds*. Abstract (PP13C-04) presented at the 2022 AGU Fall Meeting, McCormick Place Convention Center, Chicago, IL, USA.
6. **Rattanasriampaipong, R.**, Zhang, Y. G., Pearson, A., Hedlund, B., and Zhang, S. (2022, April). *Tracing ecology and evolution of marine ammonia-oxidizing archaea using archaeal lipid biomarkers*. Oral presentation, Departmental Seminar, Department of Oceanography, Texas A&M University, College Station, TX, USA.
5. **Rattanasriampaipong, R.**, Zhang, Y. G., Pearson, A., and Hedlund, B. (2021, December). *Beyond TEX₈₆: GDGTs Inform Marine Archaeal Community Ecology and Evolution*. Hybrid oral presentation at the 2021 AGU Fall Meeting, Ernest N. Morial Convention Center, New Orleans, LA, USA.
4. **Rattanasriampaipong, R.**, Zhang, Y. G., Pearson, A., Hedlund, B., and Zhang, S. (2021, September). *Beyond TEX₈₆: GDGT Distributions Inform Archaeal Ecology*. Lightning (3-minute) oral presentation at the 2021 PhanTASTIC Workshop.
3. **Rattanasriampaipong, R.** (2021, April). *Closing the gaps of Cenozoic sea surface temperature history using tetraether archaeal lipid biomarkers*. Oral presentation, Departmental Seminar, Department of Oceanography, Texas A&M University, College Station, TX, USA.
2. **Rattanasriampaipong, R.** and Zhang, Y. G. (2019, December). *Towards complete global sea surface temperature reconstructions over the Cenozoic Era*. Poster presentation at the 2019 AGU Fall Meeting, Moscone Center, San Francisco, CA, USA.
1. **Rattanasriampaipong, R.** (2013, April). *Quantification of Tsunami Magnitude from Sedimentation Modeling of Re-occurring Indian Ocean Tsunamiites at Phra Thong Island, Phang Nga, Thailand*. Oral presentation at the Annual Hitachi Senior Project Competition, Faculty of Science, Chulalongkorn University, Bangkok, Thailand.

INDUSTRY

4. **Rattanasriampaipong, R., Marksamer, A., and Kantatong, P.** (2018, April). *A new approach for pore pressure prediction using neutron-density log separation*. Oral presentation at the 2018 Sub-Surface Technical Forum, Chevron Thailand Headquarters, Bangkok, Thailand.
3. **Rattanasriampaipong, R., Paiboon, P., and Thatmali, P.** (2015, February). *Pattani Basin Regional Pore Pressure Study*. Oral presentation at the Annual Meeting of Chevron Thailand Reservoir Management Forum, Swissotel Le Concorde, Bangkok, Thailand.
2. **Rattanasriampaipong, R., Kananithikorn, N., Keawmoon, N., and Prasongtham, P.** (2014, June). *Causes of ballooning and lost circulation in Erawan and Satun fields*. Poster presentation at the Bi-Annual Meeting of Chevron Reservoir Management Forum, The Woodlands Waterway Marriott Hotel, Houston, TX, USA.
1. **Rattanasriampaipong, R., Kananithikorn, N., Keawmoon, N., and Prasongtham, P.** (2014, April). *Causes of ballooning and lost circulation in Erawan and Satun fields*. Oral presentation at the Annual Meeting of Chevron Thailand Reservoir Management Forum, Renaissance Hotel, Bangkok, Thailand.

RESEARCH MENTORING & ADVISING

UNDERGRADUATE RESEARCH MENTOR, Organic Geochemistry Laboratory, Texas A&M University 2021–2023
Trained and supervised 4 on-campus and visiting undergraduate students in laboratory preparation and analysis of archaeal lipid biomarkers (GDGTs) by LC-MS, from sample intake through data quality control:

- Freeze-drying and homogenizing marine mud samples from ocean drilling programs.
- Extracting total lipid extracts (TLE) using an Accelerated Solvent Extractor (ASE).
- Purifying samples for LC-MS analysis via cellulose filtering and silica-gel column chromatography.

GEOLOGY MENTOR, Chevron Thailand Exploration and Production Limited 2014–2015
Mentored 2 newly hired geologists on a Gulf of Thailand drilling project in wireline log interpretation, stratigraphic correlation, pore pressure prediction, and well design.

AWARDS & HONORS

AWARDS

AGU Outstanding Reviewer — Paleooceanography 2025

SCHOLARSHIPS

Texas A&M Oceanography Graduate Scholarship	2018–2024
Chevron Thailand Graduate Scholarship	2015–2016
Chevron Undergraduate Scholarship	2012–2013
Mitsui Oil Exploration Company (MOECO) Undergraduate Scholarship	2011–2013

HONORS

Invited Student Speaker, AGU Student Travel Grant Luncheon, AGU Fall Meeting, Chicago, IL, December 12, 2022.
Invited Student Speaker, Award Recognition Ceremony, Ministry of Foreign Affairs, Bangkok, June 6, 2018.
First place in Physical Sciences category, Hitachi Trophy 2013 Senior Project Competition, Faculty of Science, Chulalongkorn University, April 2013.

TRAVEL GRANTS

FAS Research and Academic Exchange, Faculty of Arts and Sciences, Harvard University	10/2024
Fall Meeting Student Travel Grant, American Geophysical Union (AGU)	12/2022
Oceanography Graduate Council (OGC) Mini Grant, TAMU Oceanography	12/2019, 12/2021, 12/2022
TAMU Research and Presentation Award, Graduate and Professional School (GPS), TAMU	12/2019

SYNERGISTIC ACTIVITIES (LEADERSHIP, SERVICE, & OUTREACH)

MANUSCRIPT REVIEWER

34 reviews (22 unique manuscripts) for 12 journals since November 2021

Journals reviewed for:

Biogeosciences (n = 1), *Chemical Geology* (2), *Climate of the Past* (1), *Frontiers in Marine Science* (2), *Folia Microbiologica* (1), *Geochimica et Cosmochimica Acta* (3), *Geophysical Research Letters* (6), *Global Planetary Change* (2), *Nature Geoscience* (1), *Organic Geochemistry* (4), *Paleoceanography and Paleoclimatology* (9), *Science Advances* (2).

SESSION CO-CONVENER AND CO-CHAIR, AGU Fall Meeting

2022–Present

Co-initiated (with Jarunetr Sae-lim), coordinated, and presided over the session *Past Climates and Environments of Southeast Asia and the Indo-Pacific (PaleoSEA)* within the Paleoclimatology and Paleoceanography (PP) section.

FOUNDER, Virtual Community of Thai Earth and Climate Scientists

2020–Present

Founded a virtual community connecting Thai graduate students and early-career researchers in the Earth and climate sciences across U.S. institutions, begun during the pandemic to share mentoring, application advice, and research talks.

INTERPRETER, Regeneron International Science and Engineering Fair (ISEF) 2026, Phoenix, Arizona

May 2026

Served as a volunteer Thai interpreter for Thai high school participants presenting their year-long science research projects at the world's largest pre-collegiate science competition.

JUDGE, Earth Week Events, The University of Arizona

Spring 2026

Served as a judge for the Climate Talk session at GeoDaze, Department of Geosciences, and evaluated undergraduate posters for the School of Earth and Environmental Sciences (SEES) poster session.

DISCUSSION LEADER, Gordon Research Seminar on Organic Geochemistry

July 27, 2024

Led the keynote session *Disciplinary Integrations with Organic Geochemistry* at the 2024 Gordon Research Seminar.

CO-AUTHOR, Open Letter to NSF on U.S. Scientific Ocean Drilling

July 8, 2022

Co-authored a letter advocating for continued NSF funding of U.S. Scientific Ocean Drilling (SciOD), which was signed by 208 Early Career Researchers from 98 institutions across 17 countries.

GRADUATE STUDENT SENATOR, Texas A&M University

2018–2022

CO-AUTHOR, GPSG Resolution on Decarbonization and Fossil Fuel Divestment

April 21, 2020

Co-authored GPSG.R.53.05, a resolution advocating for Texas A&M University to reduce fossil fuel investments, which was subsequently passed by the Graduate and Professional Student Government (GPSG) Senate.

TREASURER, Fulbright Students' Association, Texas A&M University

2019–2020

TOUR DOCENT, R/V *Sally Ride*, AGU Fall Meeting

December 11, 2019

BOARD MEMBER, Fulbright Students' Association, Texas A&M University

2018–2019

GEOLOGY UNDERGRADUATE STUDENT PRESIDENT, Chulalongkorn University

2012–2013

PRESS AND MEDIA COVERAGE

INTERVIEWED ARTICLES IN ENGLISH

1. **Brave the World** – Stories of Fulbrighters' Impact, 70th Anniversary of the Fulbright Program in Thailand. *Fulbright Thailand*, December 24, 2021. [[Online Article](#)]

INTERVIEWED ARTICLES IN THAI

5. **Learn Plearn Plearn by PTip Podcast** – Episode 190. *Fulbright Thailand*, January 23, 2024. [[Podcast](#)]
4. **Talk with Fulbrighter: Ronnakrit Rattanasriampaipong** *Fulbright Thailand*, May 18, 2022. [[Online Article](#)]
3. **Recommended Field of Study: Paleoclimatology/Paleoceanography** *Fulbright Thailand*, December 8, 2021. [[Online Article](#)]
2. **Fulbright in My View** – Fulbright Experiences. *Fulbright Thailand*, October 6, 2021. [[Online Article](#)]
1. **A Journey Without a Map** *The Science Scholars Facebook Page*, June 16, 2020. [[Online Article](#)]

PANEL DISCUSSIONS IN THAI

3. **Fulbright Thai Graduate Scholarship (TGS): More Than Just a Scholarship** – Virtual Event. *Fulbright Thailand, February 17, 2023.* [[Recording](#)]
2. **Meet the Fulbrighter Series: Study Pure & Applied Science in the U.S.** – Virtual Event. *Fulbright Thailand, March 31, 2021.* [[Recording](#)]
1. **Meet the Fulbrighter Series: Doctoral Studies in the U.S.** – Virtual Event. *Fulbright Thailand, March 5, 2021.* [[Recording](#)]

WORKSHOPS

PROFESSIONAL DEVELOPMENT

COres for Research and Education (CORE) School, Gulf Coast Repository (formerly under the International Ocean Discovery Program), Texas A&M University, College Station, TX *July 14–18, 2025*

Inaugural cohort.

Community Earth System Model (CESM) Tutorial, NCAR, Boulder, CO *July 7–11, 2025*

Lab Visit, Lab for Molecular Biogeochemistry and Organic Geochemistry (Pearson Lab), Harvard University, Cambridge, MA *October 24–30, 2024*

Hands-on experience with the Spooling-Wire Microcombustion device interfaced with an Isotope-Ratio Mass Spectrometer (SWiM-IRMS) under the guidance of Dr. Ann Pearson and PhD student Amanda Calhoun.

NOAA Climate & Global Change Summer Institute, Holiday Inn, Steamboat, CO *July 14–19, 2024*

Paleoclimate Data Assimilation Workshop, The University of Arizona, Tucson, AZ *October 18–21, 2022*

PaleoCAMP 2022 Summer School, Heising-Simons Foundation, Sierra Nevada Aquatic Research Laboratory (SNARL), Mammoth Lakes, CA *July 11–22, 2022*

Inaugural cohort. Selected as 1 of 25 attendees from 132 applicants.

Science Mission Requirements (SMR) Workshop, NSF/USSSP-IODP, Chicago, IL *May 17–18, 2022*

LinkedEarth PaleoHackathon 2, Virtual Python Workshop *October 28–29, 2021*

TEXAS A&M UNIVERSITY'S G.R.A.D. AGGIES PROFESSIONAL DEVELOPMENT PROGRAM

NSF: The Agency, Proposal Preparation & Review *February 19, 2020*

Open Educational Resources Workshop *September 13, 2019*

Creating a Life of Balance & Wellness *September 10, 2019*

FOREIGN FULBRIGHT STUDENT PROGRAM

Innovations in Civic Engagement: Harnessing Data for the Public Good, Philadelphia, PA *March 7–10, 2019*

Pre-Academic Orientation Program, Ohio University, Athens, OH *July 28–August 18, 2018*

CHEVRON IN-HOUSE TRAINING (ENERGY COMPANY TECHNOLOGY)

Reservoir Geophysics *November 21–25, 2017*

Applied Concepts of Structural Geology *June 12–16, 2017*

Stratigraphic Analysis of Shallow Marine and Fluvial Reservoirs *May 29–June 2, 2017*

Interpretation of 3D Seismic Data *November 14–17, 2016*

Basic Reservoir Engineering *December 15–19, 2014*

Formation Evaluation (Fundamentals) *October 6–9, 2014*

Applied Stratigraphic Concepts *August 4–8, 2014*

Applied Subsurface Geological Mapping *June 23–27, 2014*

Applied Petroleum Geochemistry *April 21–24, 2014*

PROFESSIONAL SOCIETIES

Member, American Geophysical Union (AGU)	2018–Present
Member, European Association of Geoscientists & Engineers (EAGE)	2025–Present
Student Member, American Association for the Advancement of Science (AAAS)	2019–2023
Student Member, American Association of Petroleum Geologists (AAPG)	2013–2023
Secretary, Thailand Society of Exploration Geophysicists (TSEG)	2015–2016
Student Member, Society of Petroleum Engineers (SPE)	2010–2013

PERSONAL INTERESTS

Science communication, Infographics, Data visualization, Marathon and ultra-distance running